

KHWAJA YUNUS ALI UNIVERSITY

School of Science and Engineering
Department of Computer Science and Engineering
Spring – 2023 Final Examination
Program: B.Sc. in CSE (Batch – 15th) 1st Year 1st Semester
Course Title: Physics - I
Course Code: PHY 1101

PART-A: MCQ

Time: 10 Minutes

Total Marks: 10x01=10

Student's ID:

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Date:

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 Invigilator's Signature:

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(These are single best type of questions. Each of the following questions or statements has five suggested answers or completions. Put tick mark (✓) on the best answer. There is no counter marking.)

Cognitive Knowledge

1. What is the normal value of form factor of AC?

- A) 0.01
- B) 0.11
- C) 1.00
- D) 1. 01
- E) 1.11

Cognitive Understanding

2. What function does **NOT** the induced emf depend on?

- A) Number turns of the coil
- B) Rate of change of magnetic flux
- C) Relative velocity between primary and secondary coil
- D) Rate of change of current flow in primary coil
- E) Resistance of the coil.

Cognitive Applying

3. How long will take to reach the peak from zero of AC signal which is supplied in our home electric line in Bangladesh?

- A) 0.005sec
- B) 0.02sec
- C) 25sec
- D) 50sec
- E) 220sec

Cognitive Understanding

4. The average value of the alternating current for full cycle, $I_{av} = ?$
[Where, I_0 is the peak value]

- A) I_0
- B) $0.707I_0$
- C) $0.637I_0$
- D) $I_0/2$
- E) 0

Cognitive Analysis

5. Two concentric circular loops of radii r_1 and r_2 carry clockwise and anticlockwise currents i_1 and i_2 . If the centre is a null point, then what will be the i_1/i_2 ?

- A) r_1/r_2
- B) r_2/r_1
- C) N_1/N_2
- D) N_2/N_1
- E) μ

Please Turn Over (P.T.O.)

Cognitive Knowledge

6. Which is the correct equation of magnetic moment?
(Where, N – no. of turns, A – area of the coil, I – current flow, B – magnetic field)
- A) $M = NIB$
 - B) $M = NAB$
 - C) $M = IB$
 - D) $M = AB$
 - E) $M = NIA$

Cognitive Knowledge

7. Which two forces are combined in Lorentz Force?
- A) Nuclear & electric
 - B) Mechanical & magnetic
 - C) Gravitational & electric
 - D) Electric & magnetic
 - E) Gravitational & magnetic

Cognitive Comprehension

8. $x = A \sin(10\pi t + \delta)$ is the solution of the differential equation of SHM:
- A) $\frac{d^2x}{dt^2} + 10\pi^2x = 0$
 - B) $\frac{d^2x}{dt^2} + 10\pi x = 0$
 - C) $\frac{d^2x}{dt^2} + 100\pi x = 0$
 - D) $\frac{d^2x}{dt^2} + 100\pi^2x = 0$
 - E) $\frac{d^2x}{dt^2} + 100x = 0$

Cognitive Understanding

9. Current flowing solenoid act as a
- A) magnet
 - B) capacitor
 - C) transistor
 - D) battery
 - E) DC generator

Cognitive Understanding

10. What is the time period of the hour hand of a clock?
- A) 24 hours
 - B) 12 hours
 - C) 1 hour
 - D) 60 sec
 - E) 0 sec

Signature of Examiner

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PART-B: SAQ

Time: 1 Hour and 50 Minutes

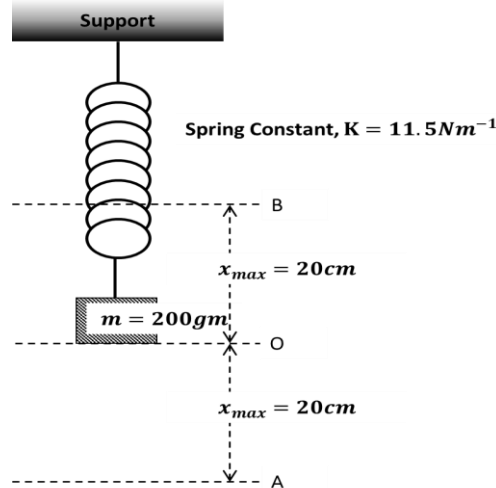
Total Marks: 5x6=30

INSTRUCTIONS: Answer question no. 01 (one) and any 04 (four) from the rest. Marks are shown by side.

Questions

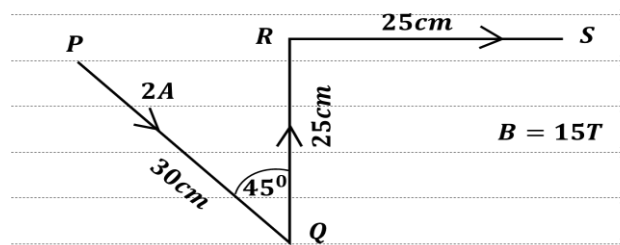
Marks

1. See the figure of a spring pendulum and answer the questions that follows:



- a) Investigate the following quantities:
- i. Angular Frequency [1]
 - ii. Velocity at Point O [1]
 - iii. Acceleration at Point B and [1]
 - iv. Total Energy. [1]
- b) Construct a wave equation for the above pendulum. [2]
2. a) Write at least 4 (four) units of magnetic field. [1]
- b) Draw the representation of the direction of magnetic field in the plane of paper. [2]
- c) Find the force of each segment of the wire: [3]

Conducting wire in a magnetic field

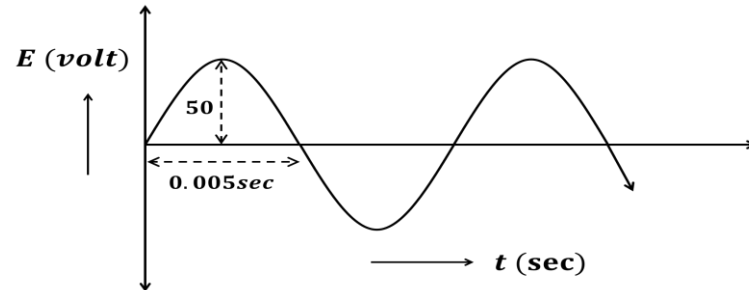


3. a) Define Self-inductance & 1 Henry. [2]
- b) Two nearby coils A and B have number of turns respectively of 200 and 1000. When 2A current flows through the coil, then magnetic flux $2.4 \times 10^{-4} \text{ wb}$ is produced in the coil A and $1.6 \times 10^{-4} \text{ wb}$ in the coil B. Calculate [4]
- (i) the self – inductance of the coil A
 - (ii) the mutual inductance of the coil B.
 - (iii) If current stops A in 0.4 sec, find the induced *emf* in coil B.

Please Turn Over (P.T.O.)

4. a) Draw a figure which can create torque in a current carrying coil in the magnetic field. [1]
 b) State Biot-Savart law. Write down the vector form of it. [2]
 c) A coil of 20 turns has an area of 800mm^2 and bears a current of 0.5A . It is placed with its plane parallel to magnetic field of intensity 0.3T . Determine the torque and magnetic moment on the coil. [3]

5. See diagram of AC current given below and answer the following question:



- a) Why is AC more dangerous than DC? [2]
 b) Calculate the peak, average, rms values & time period of above-mentioned signal. [4]
6. a) How would the attraction and repulsion develop between two parallel current-carrying coils? [1]
 b) A current of 1.5A flows through a wire 6m long. If the wire is made into a coil of 4 turns, what will be the magnetic field at its center? [2]
 c) Two long straight conductors are held parallel to each other at a distance of 0.3m . If the conductors carry currents 2A and 8A in opposite directions, then what is the distance of the neutral point from the conductor carrying smaller current? [3]